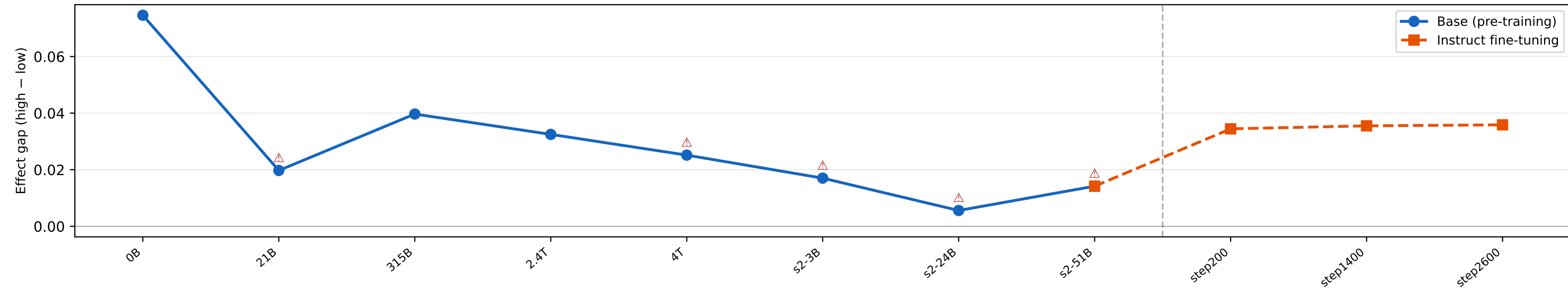


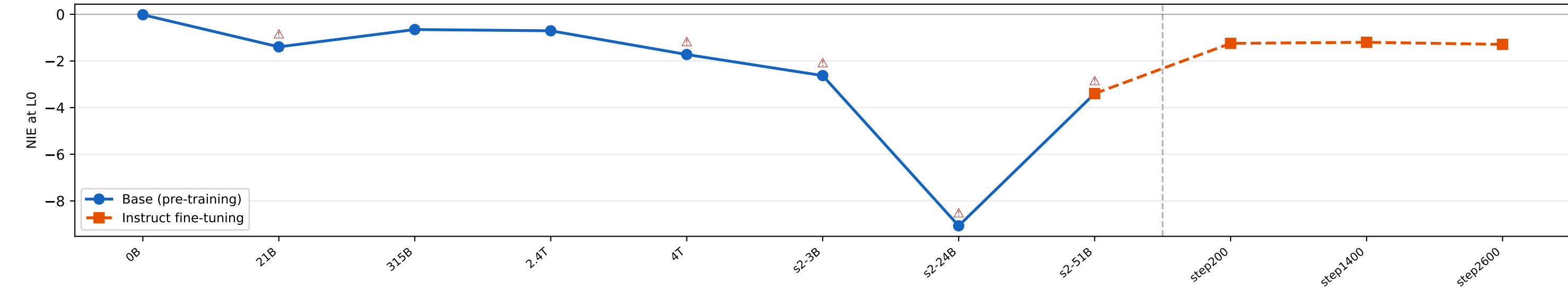
OLMo-2-0425-1B — Religion bias: learning trajectory

Left: base pre-training checkpoints | Right: instruction fine-tuning

Effect gap — how much corrupting subject tokens reduces bias-consistent probability
Larger = model relies more on subject identity for this domain



NIE at L0 — normalized indirect effect at the embedding layer
= (Patched - Corrupted) / (Clean - Corrupted)



Raw causal signal at layer 0 — absolute scale
Reflects both embedding importance and overall bias strength

